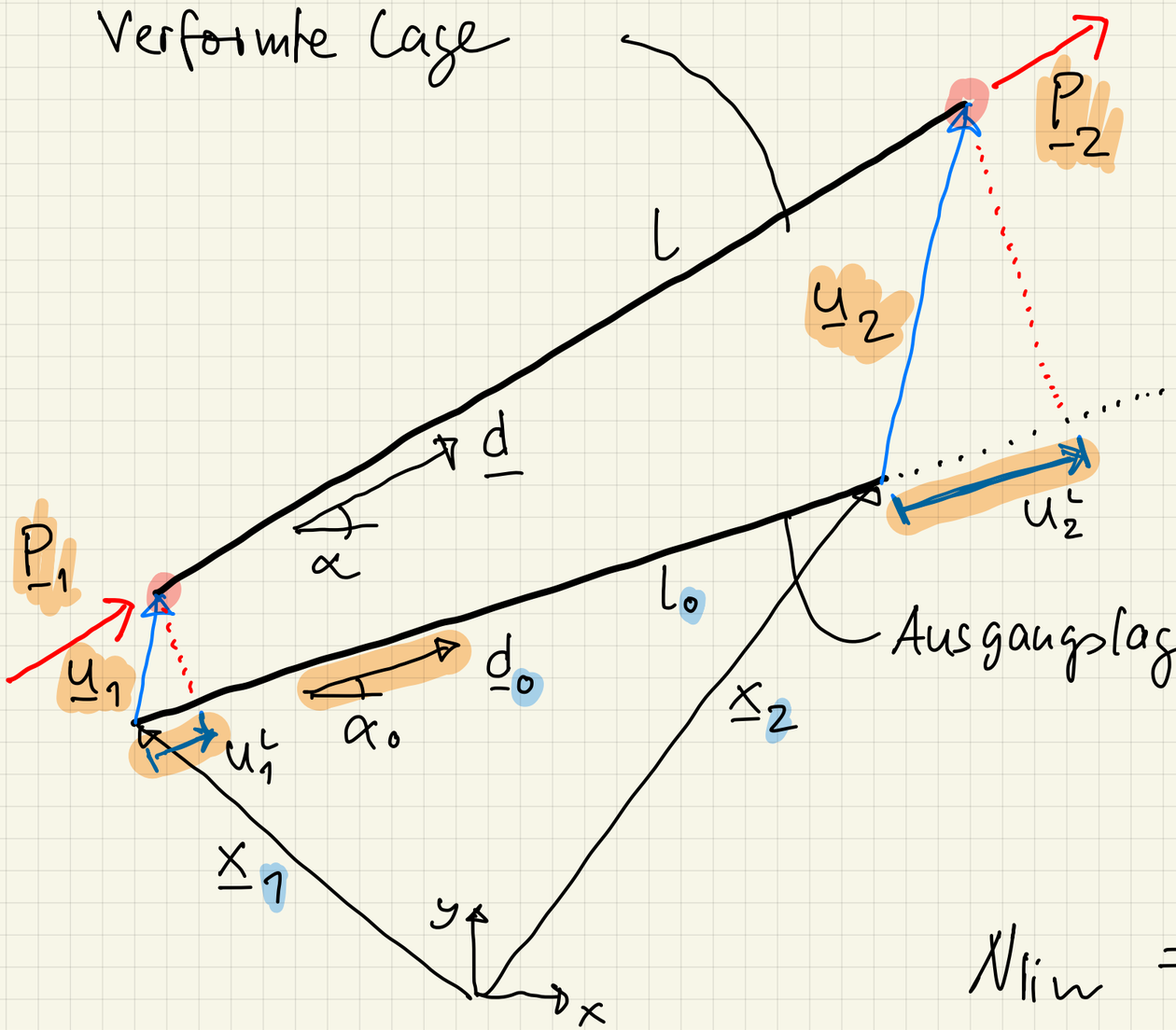
Finite Elemente für Stabtragwerke

Fachwerke : Stabendkräfte

- Formulierung für kleine Verformungen

Stabendkräfte – Formulierung für kleine Verformungen



$$\underline{t}_0 = \begin{pmatrix} -\cos\alpha_0 \\ -\sin\alpha_0 \\ \cos\alpha_0 \\ \sin\alpha_0 \end{pmatrix}$$

$$\Delta l = u_2' - u_1'$$

$$u_1' = \underline{u}_1 \cdot \underline{d}_0 = u_{11}\cdot\cos\alpha_0 + u_{12}\cdot\sin\alpha_0$$

$$u_2' = \underline{u}_2 \cdot \underline{d}_0 = u_{21}\cdot\cos\alpha_0 + u_{22}\cdot\sin\alpha_0$$

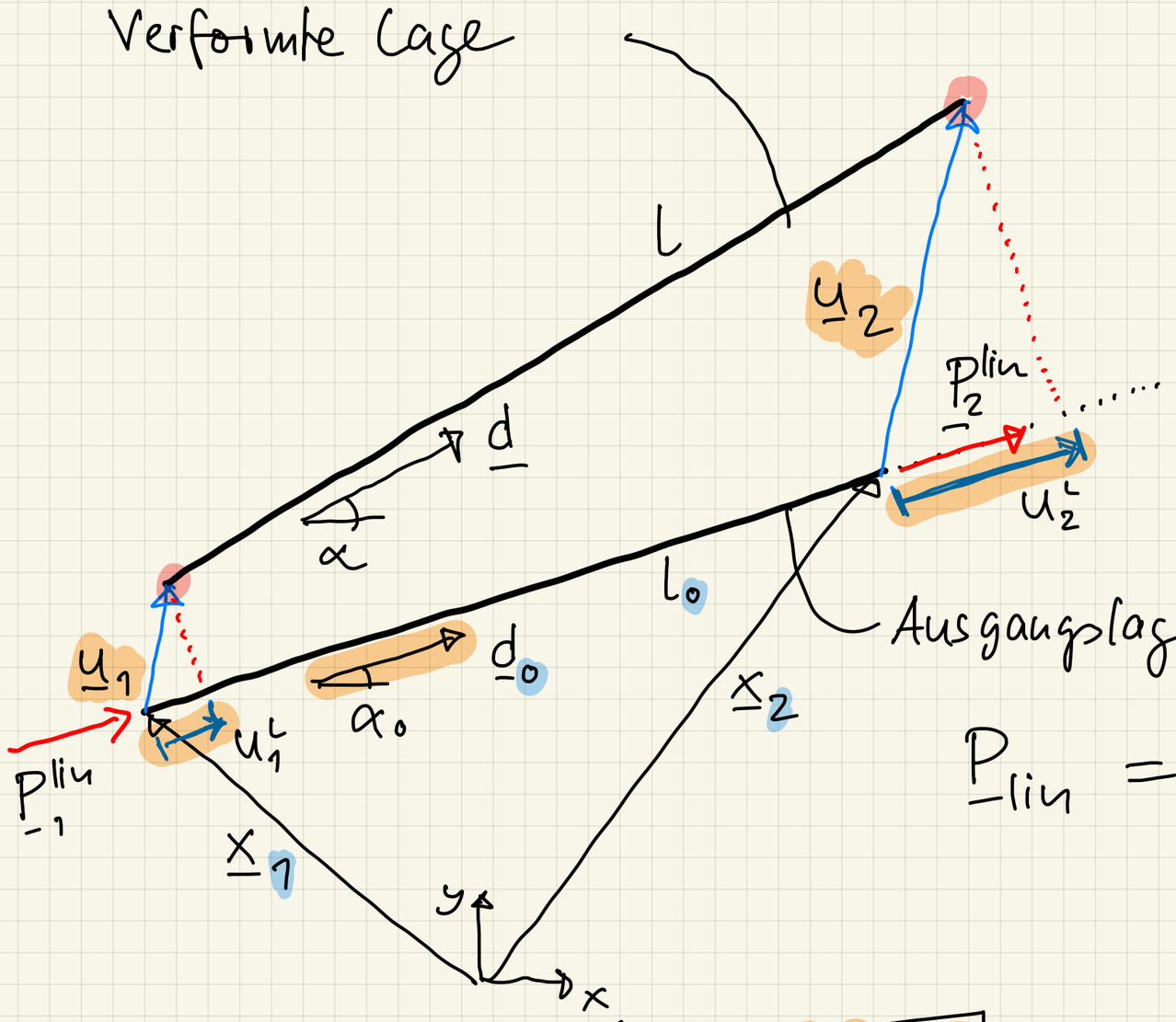
$$N_{lin} = \frac{EA}{l_0}(u_2' - u_1')$$

$$= \frac{EA}{l_0}\left(-u_{11}\cdot\cos\alpha_0 - u_{12}\cdot\sin\alpha_0 + u_{21}\cdot\cos\alpha_0 + u_{22}\cdot\sin\alpha_0\right)$$

$$N_{lin} = \frac{EA}{l_0}\begin{pmatrix} -\cos\alpha_0 & -\sin\alpha_0 & \cos\alpha_0 & \sin\alpha_0 \end{pmatrix}\begin{pmatrix} u_{11} \\ u_{12} \\ u_{21} \\ u_{22} \end{pmatrix}$$

$$\boxed{N_{lin} = \frac{EA}{l_0}\,\underline{t}_0^T\,\underline{u}}$$

Stabendkräfte – Formulierung für kleine Verformungen



$$N_{lin} = \frac{EA}{l_0}\,\underline{t}_0^T\,\underline{u}$$

$$\underline{P}_{lin} = N_{lin}\cdot\underline{t}_0 = \underline{t}_0\cdot N_{lin}$$

$$= \frac{EA}{l_0}\,\underline{t}_0\,\underline{t}_0^T\,\underline{u}$$

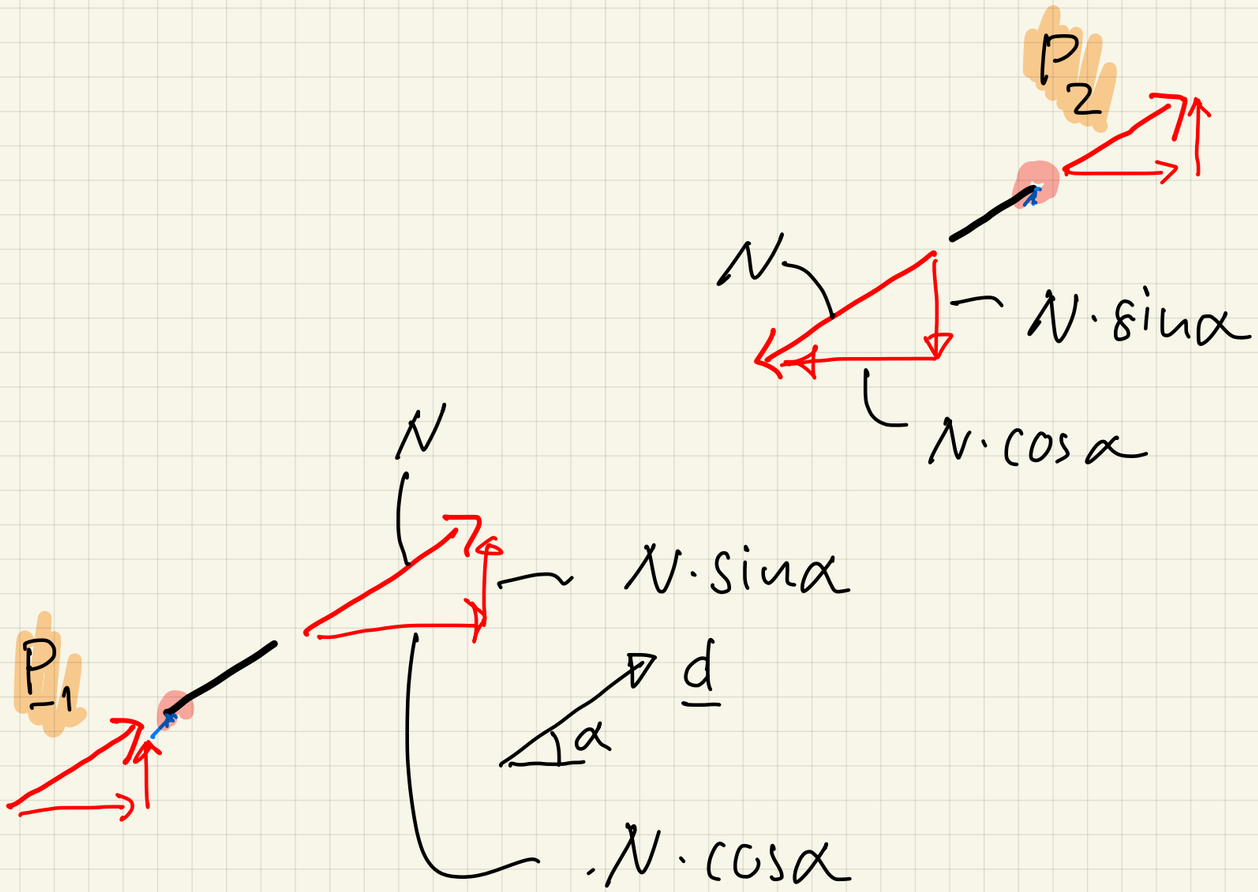
$$\underline{t}_0 = \begin{pmatrix} -\cos\alpha_0 \\ -\sin\alpha_0 \\ \cos\alpha_0 \\ \sin\alpha_0 \end{pmatrix}$$

$$\underline{P}_{lin} = \frac{EA}{l_0}\begin{pmatrix} -\cos\alpha_0 \\ -\sin\alpha_0 \\ \cos\alpha_0 \\ \sin\alpha_0 \end{pmatrix}\begin{pmatrix} -\cos\alpha_0 & -\sin\alpha_0 & \cos\alpha_0 & \sin\alpha_0 \end{pmatrix}\begin{pmatrix} u_{11} \\ u_{12} \\ u_{21} \\ u_{22} \end{pmatrix}$$

$$\underline{P}_{lin} = \underline{K}\,\underline{u}$$

$$= \underbrace{\frac{EA}{l_0}\begin{pmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{pmatrix}}_{\underline{K}}\underbrace{\begin{pmatrix} u_{11} \\ u_{12} \\ u_{21} \\ u_{22} \end{pmatrix}}_{\underline{u}}$$

$$c = \cos\alpha_0\,,\quad s = \sin\alpha_0$$



$$\underline{P}_1 = N \cdot \begin{pmatrix} -\cos\alpha \\ -\sin\alpha \end{pmatrix} = -N \cdot \underline{d}$$

$$\underline{P}_2 = N \cdot \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix} = N \cdot \underline{d}$$

$$\underline{P} = \begin{pmatrix} \underline{P}_1 \\ \underline{P}_2 \end{pmatrix} = N \begin{pmatrix} -\underline{d} \\ \underline{d} \end{pmatrix} = N \cdot \underbrace{\begin{pmatrix} -\cos\alpha \\ -\sin\alpha \\ \cos\alpha \\ \sin\alpha \end{pmatrix}}_{\underline{t}}$$

$$\underline{P} = N \cdot \underline{t} = EA \cdot \frac{l - l_0}{l_0} \cdot \underline{t}$$

